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PAPER_1018: Production Scaling v15 — 650k calc/s

Abstract

We upgrade the production benchmark from v14 (600k calc/s, 24 kernels) to v15 (650k calc/s, 30 kernels). Six new kernels cover the Session 220 physics domains:

Kernel	Description
kernel_3c273_agn_fubi	3C273 AGN F_U_Bi_i jet modulation
kernel_ton618_agn_fubi	TON618 ultramassive AGN curves
kernel_gw170817_merger	GW170817 BNS strain suppression
kernel_smbh_merger_fubi	SMBH merger inspiral F_U_Bi
kernel_dm_halo_nfw	DM halo NFW with SCm coupling
kernel_txs0506_3gamma	TXS 0506+056 3-Gamma-point profile

1. Scaling History

Version	Target (calc/s)	Kernels
v11	500,000	16
v13	550,000	20
v14	600,000	24
v15	650,000	30

2. Validation Checks

- TON618 ($2.19e-1$) > 3C273 ($1.78e-1$): Mass hierarchy confirmed
- GW170817 damped ($2.83e-23$) < undamped ($8.51e-23$): Buoyancy suppression confirmed
- All 30 kernels finite and non-negative
- Total throughput exceeds 650k target

3. REST API

22 total routes including new endpoints: - POST /api/fubi/smbh-merger — SMBH merger F_U_Bi with S26(3) - POST /api/dm/halo-nfw — DM halo NFW profile with SCm coupling

4. Implementation

File: `production_scaling_v15.py`, class `ProductionScalingV15`. CP4 class #602. Tests: 8/8 pass.